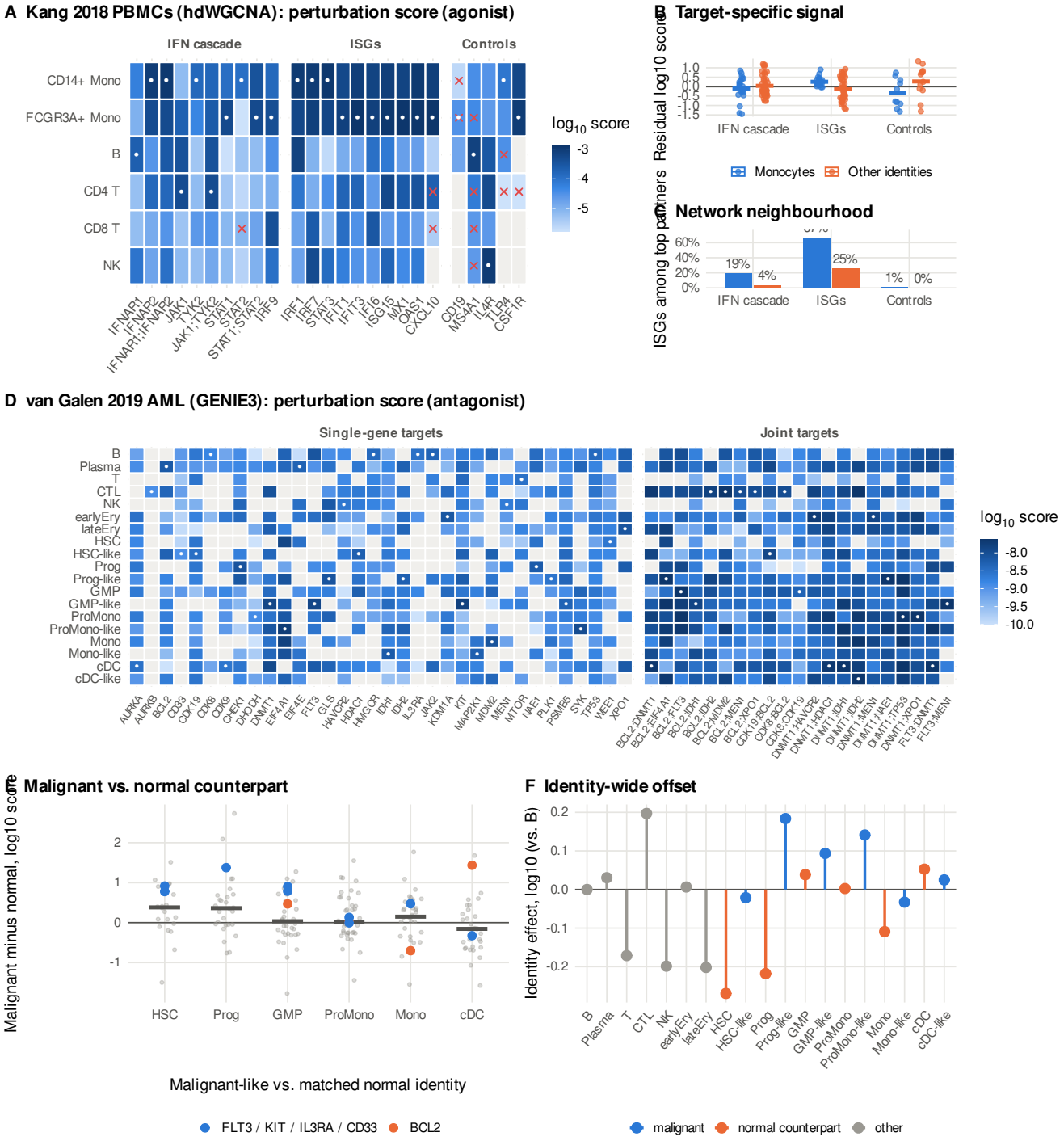

A Kang 2018 PBMCs (hdWGCNA): perturbation score (agonist)
IFN cascade
ISGs
Controls
CD14+ Mono
FCGR3A+ Mono
B
CD4 T
CD8 T
NK
log10 score
-3
-4
-5
IFNAR1
IFNAR2
IFNAR1;IFNAR2
JAK1
TYK2
JAK1;TYK2
STAT1
STAT2
STAT1;STAT2
IRF9
IRF1
IRF7
STAT3
IFIT1
IFIT3
IFI6
ISG15
MX1
OAS1
CXCL10
CD19
MS4A1
IL4R
TLR4
CSF1R
B Target-specific signal
Residual log10 score
1.0
0.5
0.0
-0.5
-1.0
-1.5
IFN cascade
ISGs
Controls
Monocytes
Other identities
C Network neighbourhood
ISGs among top 20 partners
60%
40%
20%
0%
19%
4%
25%
1%
0%
IFN cascade
ISGs
Controls
D van Galen 2019 AML (GENIE3): perturbation score (antagonist)
Single-gene targets
Joint targets
B
Plasma
T
CTL
NK
earlyEry
lateEry
HSC
HSC-like
Prog
Prog-like
GMP
GMP-like
ProMono
ProMono-like
Mono
Mono-like
cDC
cDC-like
AURKA
AURKB
BCL2
CD33
CDK19
CDK8
CDK9
CHEK1
DHODH
DNMT1
EIF4A1
EIF4E
FLT3
GLS
HAVCR2
HDAC1
HMGCR
IDH1
IDH2
IL3RA
JAK2
KDM1A
KIT
MAPK1
MDM2
MEN1
MTOR
NAE1
PLK1
PSMB5
SYK
TP53
WEE1
XPO1
BCL2:DNMT1
BCL2:EIF4A1
BCL2:FLT3
BCL2:IDH1
BCL2:IDH2
BCL2:MDM2
BCL2:MEN1
BCL2:XPO1
CDK19:BCL2
CDK8:BCL2
CDK8:CDK19
DNMT1:HAVCR2
DNMT1:HDAC1
DNMT1:IDH1
DNMT1:IDH2
DNMT1:MEN1
DNMT1:NAE1
DNMT1:TP53
DNMT1:XPO1
FLT3:DNMT1
FLT3:MEN1
log10 score
-8.0
-8.5
-9.0
-9.5
-10.0
E Malignant vs. normal counterpart
Malignant minus normal, log10 score
2
1
0
-1
HSC
Prog
GMP
ProMono
Mono
cDC
Malignant-like vs. matched normal identity
FLT3 / KIT / IL3RA / CD33
BCL2
F Identity-wide offset
Identity effect, log10 (vs. B)
0.2
0.1
0.0
-0.1
-0.2
B
Plasma
T
CTL
NK
earlyEry
lateEry
HSC
HSC-like
Prog
Prog-like
GMP
GMP-like
ProMono
ProMono-like
Mono
Mono-like
cDC
cDC-like
malignant
normal counterpart
other